

Comparative Analysis of the Comovement of CDS spreads, Stock Returns and Implied Volatility across High and Low Economic Uncertainty Regimes

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1 Abstract

Credit Default Swaps are a key instrument in modern financial markets, used by financial agents to hedge against default risk, lower the risk of lending, and increase their returns on capital. The par spread of a CDS is often used as an indication of the health of the underlying security, while stock returns and implied volatility summarize conditions in the underlying equity market. We study the links between credit and equity markets by analyzing the comovement of the aforementioned variables. We divide the market into periods of high and low Economic Policy Uncertainty and conduct a comparative study using a Threshold Vector Autoregression (TVAR). Moreover, we partition the market and conduct sectoral analysis on over 220 securities. We find that there is a strong bidirectional Granger Causality in the High and Low Uncertainty Regimes. However, the strong linkage between the variables collapses when the market is partitioned into sectors.

2 Introduction

The COVID-19 pandemic was instrumental in influencing public opinion on systemic risk in the US financial market. Byun, Kim, and Oh (2025) find that *systemic credit risk premia* (premium demanded by investors to hold onto assets susceptible to multiple defaults) saw a steady decline between 2016 and 2019, followed by a spike in 2020, which later stabilized at a higher level than the pre-COVID baseline. The Economic Policy Uncertainty Index by Baker, Bloom and Davis (2016) also observes a shock in 2020 and later stabilises until 2024. In financial markets, negative market sentiment result in uncertainty about firm value and consequently increased volatility (Black, 1986) and default risk (Merton, 1974).

Credit Default Swaps are financial instruments that allow for the transfer of credit risk between buyers and sellers in the market. The buyer pays periodic premiums to the seller, who agrees to compensate the

buyer if a specified credit event (default, restructuring, etc.) occurs on the reference entity. The CDS spread represents the running premium required to buy this protection. It is calculated as the running premium that sets the contract's present value to zero.

This paper aims to study the determinants of CDS spreads through their comovement with stock returns and implied volatility. Our goal is to compare these dynamics in both low (Low EPU) high (High EPU) periods. Dividing our analysis into a high and low regime helps us understand the degree and direction of these relationships during periods of differing uncertainty.

Understanding how equity, credit, and option markets interact under policy stress is essential for regulators, investors, and risk managers. Although Da Fonseca and Gottschalk (2013) have documented time-varying linkages between CDS, equity returns, and volatility, they haven't considered macro-level shocks in their analysis. Moreover, much of the existing literature relies on historical (realized) volatility as the primary measure of risk. By construction, realized volatility is purely backward-looking and therefore inherently limited in a forecasting context.

3 Literature Review

Single-name CDS spreads are widely interpreted as forward-looking measures of firm-level risk. Through arbitrage relationships between corporate bonds and risk-free instruments, CDS spreads encapsulate the market's assessment of default probabilities and expected loss.

Blanco, Brennan and Marsh (2004) find evidence that information about firm credit risk shows up first in CDS markets, and that credit spreads are positively correlated with firm default probability. In Merton's (1974) framework, the value and volatility of a firm's assets map directly into the pricing of corporate debt and credit spreads. This foundation, along with linkages between equity risk signals and corporate bonds outlined in Campbell and Taksler (2003), provides the theoretical foundation for using equity market signals and implied volatility to explain CDS spreads.

Several empirical papers find stock price returns and volatility to be key determinants of CDS spreads. Galil et al. (2014) compare a wide set of explanatory variables and identify stock returns and changes in 250-day annualized stock return volatility as among the strongest predictors of contemporaneous CDS spread changes, though the overall structural model explains a relatively small amount of variance. This lends to the possibility of market effects playing a substantial role in pricing. This paper serves as a foundation for our tri-variable structural model, though we opt for implied volatility, as backward-looking measures smooth over prolonged periods and thus do not allow for analysis surrounding daily shocks.

The importance of option-implied volatility and market expectations is outlined in relevant literature.

Benkert (2004) shows that implied volatility often carries information about credit risk beyond historical (realized) volatility. Furthermore, Tang and Yan (2010) identify implied volatility as a significant firm-level determinant of credit spreads and document the significant interaction between market conditions, volatility, and credit spreads. Duong, Park (2022) provide further evidence that the CDS market captures forward-looking information from equity options markets. Additionally, at-the-money implied volatility is found to be a robust determinant of CDS returns across firms on the same day. This informs our variable selection.

To analyze the effects of high-frequency daily macro shocks, we turn to time-series literature that uses vector autoregressive (VAR) frameworks and Granger causality tests to evaluate dynamic, lagged interactions among financial variables. Granger’s (1969) formulation provides the econometric foundation for testing whether past values of one series contain information useful for forecasting another. VARs have been widely applied in the analysis of credit markets. Srivastava (2016), Billio (2010), and Da Fonseca, Gottschalk (2020) utilize VAR models and Granger tests to explore predictive financial relationships. Building on single regime VAR models, Tong(1991) and Hansen (1999) outline the Threshold VAR methodology for evaluating predictive relationships across threshold-defined regimes and discusses its use cases for nonlinear time-series data. Alfonso, Baxa, and Slavic (2011) provide an empirical example of using an exogenous market stress indicator as a threshold variable among financial variables for TVAR analysis.

The introduction of macroeconomic shocks into our structural model is motivated in part by Kaviani et al. (2020), who find evidence of the effects of policy uncertainty on firms and credit markets. The Economic Policy Uncertainty Index, introduced in Bloom, Baker and Davis (2016), creates and defines use cases for an index that tracks newspaper semantics. Beyond identifying the correlation between their index and known economic crises, their research employs vector autoregressions to measure the effects of EPU shocks on stock market volatility, among other variables. The introduction of an exogenous macroeconomic variable, not mechanically correlated with structural variables, allows us to test for indirect shocks to our structural variables. Tang and Yan (2010) also identify investor sentiment, which can be linked to media/journalistic output, and implied volatility, as the most significant determinant of CDS spreads. Our analysis builds on the literature by combining an exogenous measure of media sentiment with high-frequency (daily) market movements using implied volatility. Time series analysis on market and sector aggregates allows us to focus on market-wide shock effects, rather than backward-facing firm-level determinants.

4 Data

Our implied volatility data was obtained from OptionMetrics IvyDB. For each trading-day, we select the options contract with the highest trading volume to ensure maximal liquidity and price reliability. When

multiple contracts share the highest volume, we select the contract whose option delta is closest to 0.50/-0.50, corresponding to a near at-the-money option. Our firm-level price returns and CDS spread were sourced from CRSP and Markit, respectively. Price data was taken from firms on the NASDAQ and the New York Stock Exchange. CDS data was taken only for contracts on United States firms and with a 5Y tenor, as these are most liquid.

We compiled our dataset by filtering only for firm tickers that were common across all three datasets and remained throughout our entire seven-year period. We do this to avoid bias from companies with missing values or that were delisted for failing to meet requirements, financial distress, etc. The telecommunications sector was dropped due to a lack of firms (N=2). The summary statistics reveal a small number of extreme observations in daily changes in implied volatility, reflecting rare but economically meaningful episodes of sudden uncertainty. All financial variables enter the model as differences or returns from the previous market day, providing zero mean stationarity while allowing the VAR and TVAR systems to capture the dynamic effects of shocks. The summary statistics for the primary variables can be seen in Table 1.

Finally, Economic Policy Uncertainty (EPU), defined in Bloom (2009), is used as our macroeconomic shock proxy. The United States Daily EPU is calculated using the prevalence of articles in US newspapers that contain at least one term relating to the economy, uncertainty and policy. The index is normalized using the total number of articles. As shown by Baker, Bloom and Davis (2016) and Dai, Xiong and Zhou (2021), the EPU is shown to be correlated with economic events, market volatility and stock market strength, and is validated as an uncertainty proxy. EPU is selected to avoid other common market-temperature proxies such as the CBOE Volatility Index and the CNN Fear and Greed Index, which would introduce mechanical endogeneity issues as they are constructed from the same underlying option prices and equity data used in our primary model.

Table 1: Summary Statistics of Key Variables

Variable	N	Mean	SD	Min	Max
Stock Returns	292 086	0.000 69	0.023 21	−0.483 26	0.878 48
Policy Uncertainty (EPU)	292 086	134.961 33	97.443 00	16.750 00	807.660 00
Δ Implied Volatility	292 086	0.000 01	0.101 05	−6.358 57	7.713 69
Δ CDS Spread	292 086	0.000 00	0.001 21	−0.120 08	0.104 53

5 Methodology

5.1 Model Variables

Change in Implied Volatility: The implied Volatility of a security is the market-implied variability of its future stock price estimated through its options prices. Implied volatility is a forward-looking measure, as opposed to realized volatility, whose co-movement with CDS Spread has been studied by Da Fonseca and Gottschalk (2013). We endeavour to gain insight on how change in implied volatility influences and is influenced by our other model variables.

Returns: The return in our dataset is given by the change in the daily closing price of the underlying stock. This is not to be confused with the return on equity which is another factor shown to affect CDS Spread (Duong and Park, 2018).

CDS Parspread: CDS Par Spread is largely determined by its market-implied term structure. The term structure is dependent on an associated hazard curve that gives the probability of default with time. Lower returns signal less market interest in said stock and thus imply higher CDS premiums. Similarly, higher volatility signals increased risk associated with the underlying commodity and thus increases CDS spreads. However, with time series data, causal claims are sparsely substantiated, particularly since most of the market is white noise. Thus, our interest lies in the predictive power of our primary variables on the change in CDS premiums. This can be done through Granger Causality.

To justify the methodology used in our paper, we provide a concise overview of the mathematics behind Vector Autoregressive (VAR) models, Granger Causality and Threshold VARs.

5.2 VAR Models

5.2.1 Building VAR(p)

Let us consider the time series given by:

$$y_{i,t+h} = f_i(y_{1,t-1}, y_{2,t-1}, \dots, y_{K,t-1}, \dots, y_{K,t-1}, \dots, y_{K,0}) \text{ where } i \in 1, \dots, K$$

Assumption 1: Our time series f_i is linear (this gives us a vector autoregressive model). For example in the case of a time series with one lag:

$$\begin{aligned} y_{i,t} &= f_i(y_{1,t-1}, y_{2,t-1}, \dots, y_{K,t-1}, \dots, y_{1,0}, \dots, y_{K,0}) = f_i(y_{1,t-1}, y_{2,t-1}, \dots, y_{K,t-1}) \\ \implies y_1 &= v + Ay_0 + u_0, \quad y_2 = v + Ay_1 + u_1, \text{ where } y_0, v, u_1 \in \mathbb{R}^K \text{ and } A \text{ a } K \times K \text{ matrix.} \\ y_2 &= v + A(v_0 + Ay_0 + u_0) + u_1 = (I_k + A)v + A^2 + (Au_0 + I_k u_1) \end{aligned}$$

Repeating this process, we get:

$$y_t = \sum_{i=0}^{t-1} A^i v + A^t y_0 + \sum_{j=0}^{t-1} A^j u_{t-j}.$$

Assumption 2: VAR process is started in $t = -\infty$. This simplifying assumption is particularly easy to justify in our model, given the fact that all our variables extend the entire 7 year period.

Assumption 3 (Stability and Stationarity): The process is stable and consequently does not diverge (all of its eigenvalues are less than 1). Through 2 and 3, we get:

$$\implies y_t = \mu + \sum_{i=1}^{\infty} A^i u_{t-i}, \text{ where } \mu = (I_K - A)^{-1}v.$$

Assumption 4 (White Noise): u_i is white noise with expected value 0. Moreover, we expect this variable to be distributed log-normally.

$$\implies E(y_t) = \mu \forall t \text{ and } \Gamma_y(h) = E[(y_t - \mu)(y_{t-h} - \mu)'] = \sum_{i=0}^{\infty} A_1^{h+i} \Sigma_u A_1^{i'}.$$

We now expand our model to a process generated by p lags. This can be done by considering the block vectors and matrices:

$$Y_t := \begin{bmatrix} y_t \\ y_{t-1} \\ \vdots \\ y_{t-p+1} \end{bmatrix}_{(Kp \times 1)}, \quad \nu := \begin{bmatrix} \nu \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{(Kp \times 1)},$$

$$A := \begin{bmatrix} A_1 & A_2 & \cdots & A_{p-1} & A_p \\ I_K & 0 & \cdots & 0 & 0 \\ 0 & I_K & & 0 & 0 \\ \vdots & & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & I_K & 0 \end{bmatrix}_{(Kp \times Kp)}, \quad U_t := \begin{bmatrix} u_t \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{(Kp \times 1)}.$$

$$Y_t = \nu + \mathbf{A}Y_{t-1} + U_t$$

Here $\mu = E(Y_t) = (I_{Kp} - \mathbf{A})^{-1}\nu$ and $\Gamma_y(h) = E[(y_t - \mu)(y_{t-h} - \mu)'] = \sum_{i=0}^{\infty} \mathbf{A}_1^{h+i} \Sigma_u \mathbf{A}_1^{i'}$.

The values for the matrices A and vector ν are calculated using OLS. Moreover, given that we want to forecast h periods into the future, we find an estimate such that they minimize the Mean Square Error. This is given by $E_t(Y_{t+h}) = E(Y_{t+h}|Y_t)$ (Lütkepohl 2007). These forecasts are calculated recursively.

5.3 Granger Causality

Given a VAR(p) process, we can pick from Y_t two processes x_t and y_t . A process x_t Granger Causes y_t if it makes our forecast of y_t more efficient. Let Ω_t and $\Omega_{x,t}$ be the information provided to us at time t , and the information provided to us at time t by x_t . We say x_t Granger Causes y if:

$$MSE_y(h|\Omega_t) \leq MSE_y(h|\Omega_t - \Omega_{x,t}) \text{ for some } h$$

Granger Causality is not a proof of a causal relationship between our variables however it indicates the explanatory power of change in CDS Parspreads, change in Implied Volatility and Stock Returns on each other. We test Granger Causality through the Wald Test.

For a two-variable VAR(p) where cds_t is the process that generates the change in average CDS Parspread for a portfolio and VR_t is the process that generates the change in Implied Volatility of a portfolio and Returns from the portfolio:

$$y_t = \begin{pmatrix} cds_t \\ VR_t \end{pmatrix} = \nu + A_1 y_{t-1} + \dots + A_p y_{t-p} + u_t,$$

$$A_k = \begin{pmatrix} A_{11}^{(k)} & A_{12}^{(k)} \\ A_{21}^{(k)} & A_{22}^{(k)} \end{pmatrix}.$$

Testing whether VR Granger-causes CDS corresponds to:

$$H_0 : a_{12}^{(1)} = \dots = a_{12}^{(p)} = 0.$$

We consider all the coefficients given by the matrix A_k above

$$\beta = \left(\nu_1, a_{11}^{(1)}, \dots, a_{11}^{(p)}, a_{12}^{(1)}, \dots, a_{12}^{(p)}, \nu_2, a_{21}^{(1)}, \dots, a_{21}^{(p)}, a_{22}^{(1)}, \dots, a_{22}^{(p)} \right)'.$$

This vector has $4p + 2$ elements. To test non-causality from VR to CDS , select the p coefficients of VR_{t-k} in the first equation:

$$H_0 : C\beta = 0,$$

where

$$C = \begin{pmatrix} 0_{p \times (p+1)} & I_p & 0_{p \times (2p+1)} \end{pmatrix}$$

$$C = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \text{ (when } p = 2)$$

Let $\hat{\beta}$ be the OLS estimate. Let Z be the regressor matrix and $\hat{\Sigma}_u$ the estimated residual covariance matrix. Then the Wald statistic is:

$$\lambda_W = (C\hat{\beta})' \left[C \left((ZZ')^{-1} \otimes \hat{\Sigma}_u \right) C' \right]^{-1} (C\hat{\beta}). \text{ (Lütkepohl 2007)}$$

Under H_0 :

$$\lambda_W \xrightarrow{d} \chi^2(p).$$

A finite-sample F -version is used in our case:

$$\lambda_F = \frac{\lambda_W}{p} \sim F(p, d),$$

where d is the degrees of freedom in the equation.

5.4 Threshold Vector Autoregression

Following our baseline analysis of lead-lag relationships using Granger causality, we expand our framework to capture non-linearities in the co-movement of CDS spreads, implied volatility, and price returns. We introduce the Economic Policy Uncertainty (EPU) index as our threshold variable, allowing the model to switch regimes endogenously based on the level of policy uncertainty.

For a sample of $t \in \{1, 2, \dots, T\}$ observations, let y_t denote the vector of our three endogenous variables:

$$y_t = \begin{pmatrix} \Delta(CDS_t) \\ returns_t \\ \Delta Volatility_{imp,t} \end{pmatrix}$$

Our two-regime TVAR model is defined as a piecewise linear system dependent on the state of the running threshold variable EPU_t relative to a threshold value τ , as defined in Hansen (1999) and Tsay (1998):

$$y_t = \begin{cases} \beta_{1,0} + \beta_{1,1}y_{t-1} + \dots + \beta_{1,p}y_{t-p} + \varepsilon_{1,t} & \text{for } EPU_t \leq \tau \quad (\text{Regime 1}) \\ \beta_{2,0} + \beta_{2,1}y_{t-1} + \dots + \beta_{2,p}y_{t-p} + \varepsilon_{2,t} & \text{for } EPU_t > \tau \quad (\text{Regime 2}) \end{cases}$$

Where $\beta_{r,0}$ is a 3×1 intercept vector specific to regime r , and $\beta_{r,j}$ are 3×3 coefficient matrices representing the transmission mechanism of the variables at the j -th day lag within regime r . The threshold τ is determined endogenously in each model by minimizing the Sum of Squared Residuals (SSR) across regimes 1 and 2:

$$\tau = \underset{\tau}{\operatorname{argmin}} (SSR_1(\tau) + SSR_2(\tau))$$

For a given regime r , we define the stacked regressor vector x_t with dimension $(1 + 3p) \times 1$ as $x_t = \begin{pmatrix} 1 & y'_{t-1} & \dots & y'_{t-p} \end{pmatrix}'$. If for each regime, we let X_r stack all x_t , and Y_r stack all y_t , then our estimator matrix becomes:

$$\widehat{B}_r = (X_r' X_r)^{-1} (X_r' Y_r)$$

where every coefficient matrix for our 3 variables $\beta_{r,j}$ is extracted from the rows of $\widehat{B}_r$. Each element of $\beta_{r,j}$ measures the marginal predictive effect of the j -day lag of one variable on the dependent variable, conditional on all other lags, variables, and the firms being in the regime r .

To run our regressions at both the market and sectoral levels, we first formulate equal-weighted portfolios. By running time series analysis on the cross-section, we assume perfect correlation across firms.

5.5 Portfolio Construction

Our raw data consists of firm level cds spread, implied volatility and stock returns data from 2017 to 2023 across 220 Firms in the United States. Running our TVAR model and testing for Granger Causality in the low and high EPU regime was computationally out of our scope. Moreover we realized it is more effective to construct market wide and sectoral equal weighted portfolios, for which we could run the time series on. Let us consider a portfolio $S \in \mathbf{N}$ where S is a set of firms selected from $\mathbf{N}$ (the set of all the firms).

$$Returns_{s,t} = Price_{s,t} - Price_{s,t-1} \quad s \in S$$

For an equal weighted portfolio:

$$Price_{S,t} = \frac{1}{|S|} \sum_{s \in S} Price_{s,t} \implies \\ Returns_{S,t} = \frac{1}{|S|} \sum_{s \in S} Price_{s,t} - \frac{1}{|S|} \sum_{s \in S} Price_{s,t-1} = \frac{1}{|S|} \sum_{s \in S} Returns_{s,t}.$$

Thus average returns of the constituents of the portfolio were computed in the case of portfolio returns. The same can be done for CDS par spread values (a portfolio where equals swaps for each underlying security had been bought).

$$CDS_{S,t} = \frac{1}{|S|} \sum_{s \in S} CDS_{s,t}$$

The same averaging trick doesn't apply to Implied Volatility.

$$IV_{S,t} = \mathbb{E}(Var(Price_{S,t})) = \mathbb{E}(Var(\frac{1}{|S|} \sum_{s \in S} Price_{s,t})) = \frac{1}{|S|^2} \mathbb{E}(Var(\sum_{s \in S} Price_{s,t})).$$

The portfolio implied volatility is now determined through a covariance matrix. Particularly using the formula:

$$IV_{S,t} = \sqrt{\mathbf{w}^\top \Sigma_S \mathbf{w}}$$

where $\mathbf{w} = \left(\frac{1}{|S|}, \frac{1}{|S|}, \dots, \frac{1}{|S|} \right)^\top$ is the equal-weight vector. In expanded form, the $|S| \times |S|$ covariance matrix is:

$$\Sigma_S = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \sigma_{13} & \cdots & \sigma_{1n} \\ \sigma_{21} & \sigma_2^2 & \sigma_{23} & \cdots & \sigma_{2n} \\ \sigma_{31} & \sigma_{32} & \sigma_3^2 & \cdots & \sigma_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \sigma_{n3} & \cdots & \sigma_n^2 \end{pmatrix}$$

which simplifies to:

$$IV_{S,t} = \frac{1}{|S|} \sqrt{\sum_{i=1}^n \sigma_i^2 + 2 \sum_{i=1}^{n-1} \sum_{j=i+1}^n \sigma_{ij}}$$

This was computed using a $|S| \times |S|$ covariance matrix on R. 10 such portfolios were constructed: one for the entire market $S = \mathbf{N}$ and one for each of our 9 sectors.

6 Diagnostics

6.1 Model Diagnostics

6.1.1 Augmented Dick-Fuller Test for Variable Stationarity

The Augmented Dickey-Fuller (ADF) test examines whether a time series has a unit root (non-stationary). The test determine whether the input time series (cds_t , iv_t , ret_t) has a first and second moments that are independent of time t (if they don't we can't construct a VAR as demonstrated in 5.2). The test estimates the regression:

$$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{i=1}^p \delta_i \Delta y_{t-i} + \varepsilon_t,$$

The null hypothesis is $H_0 : \gamma = 0$, indicating a unit root and non-stationarity. The alternative is $H_1 : \gamma < 0$, implying stationarity. All of our variables passed the test.

6.1.2 Choice of Lag Coefficients

The choice of lag coefficients is integral when running a time series model. Choosing too many lags results in overfitting of the model (model becomes optimized for our dataset but loses any explanatory power) and choosing too few leads to omission of key variables. The lag coefficient is chosen through Akaike Information Criteria, Bayesian Information Criteria (Schwarz Criteria), Hannan-Quinn Criteria or Final Prediction Error. Given our parameters and objective of getting a consistent estimate of our variables to run the Granger Causality test, we chose a lag of two as suggested by the Schwarz Criteria. Schwarz Criteria provides the most consistent estimate when a set of standard assumptions is applied on each model (Lutkepohl, 2007).

6.2 Model Output Diagnostics

6.2.1 Eigenvalue Decomposition Test for Stability

To ensure the validity of our VAR model, we conduct an eigenvalue decomposition test for stability. For the VAR(p) processes that we get, stability requires that all eigenvalues of the companion matrix have modulus less than one. The companion matrix is constructed from the coefficient matrices as:

$$A = \begin{bmatrix} A_1 & A_2 & \cdots & A_{p-1} & A_p \\ I_K & 0 & \cdots & 0 & 0 \\ 0 & I_K & \ddots & \vdots & \vdots \\ \vdots & \ddots & \ddots & 0 & 0 \\ 0 & \cdots & 0 & I_K & 0 \end{bmatrix},$$

where A_i are the estimated coefficient matrices for time i and I_K is the identity matrix of dimension K . We compute these eigenvalues and verify they have modulus less than one making our final model is suitable for further analysis.

6.2.2 Portmanteau Test for Autocorrelation

In the context of Vector Autoregressive (VAR) models, the coefficient of determination R^2 is an inadequate goodness-of-fit measure. This is because R^2 increases with the addition of lags, even if those lags do not improve the model's true explanatory power. This property encourages overfitting and provides no penalty for model complexity (Lutkepohl 2007). Moreover R^2 does not account for the dynamic interdependence among variables in a multivariate system, focusing only on individual equation fit (Hamilton 1994). Lastly, R^2 cannot detect misspecification in the form of residual autocorrelation—a key violation of the VAR assumption that errors are serially uncorrelated (Lutkepohl 2007). Therefore, to analyse the validity of our results we employ the Portmanteau Test.

The Portmanteau test evaluates whether the residuals from a VAR model exhibit serial correlation, which would indicate model misspecification. The core idea is to examine whether residual autocorrelation matrices across multiple lags are jointly zero. If residuals are truly white noise (as required for valid inference), their autocorrelations should be negligible. The test aggregates information from residual autocorrelations up to a specified lag h into a single test statistic.

The null hypothesis states that all residual autocorrelation matrices from lag 1 to lag h are zero:

$$H_0 : \mathbf{R}_h = (R_1, \dots, R_h) = 0,$$

where R_i is the $K \times K$ residual autocorrelation matrix at lag i . Under the null hypothesis and standard regularity conditions, the test statistic follows an approximate χ^2 distribution.

The results of the Portmanteau Test indicate a very low likelihood of serial correlation in our Market Portfolio for both High EPU and Low EPU periods. This indicates that there is a very low likelihood of there being dynamics between these three variables that haven't been considered in the model. All sector portfolios with the exception of Financials (Low EPU), Industrials (High EPU) and Technology (High EPU) pass the Portmanteau Test. This indicates missed dynamics in these VARs. Industrials in particular sees

high p values in both High and Low EPU periods indicating some autocorrelation. This may indicate some systemic error in our data or dynamics missed by our model.

7 Empirical Analysis

7.1 Market Portfolio Analysis

By running a preliminary vector autoregressive model over our entire market and time period, we establish baseline predictive relationships. Using two-day lags we find statistically significant lead-lag relationship coefficients for both implied volatility and returns to CDS spreads [Table 2]. This relationship is bidirectional for returns, with CDS spreads also leading returns in both 1-day and 2-day lags. However, volatility is led only by returns. These results align with structural credit-risk models (Merton 1974) and with empirical literature showing tight volatility-credit linkages (Zhang, Zhou, Zhu 2005) and strong effects of volatility in the variation of CDS spreads (Campbell and Taksler 2003). Our Granger causality tests pass with highly statistically significant p -values to reject the null in all directions, presenting evidence of dynamic dependence across the stock and credit markets [Table 3]. The statistical significance of the relationships also implies a feedback system between the variables, rather than a chain of transmission from one to another. In [Table 3], Granger causality tests are conducted system-wide; each reported direction reflects a joint F-test of whether the lagged values of the predictive variable significantly improve forecasts of the other endogenous variables in the VAR.

To enhance our understanding of the regime importance of market-wide uncertainty, we continue to our TVAR model. Using a trim we ensure at least 15% of observations are in each regime. The endogenously determined EPU threshold value was 197.2. In the low-uncertainty regime, both lagged equity returns and lagged changes in implied volatility exert statistically significant effects on CDS spread changes. Specifically, a one-unit increase in lagged implied volatility changes is associated with a 0.0018-unit increase in next-day CDS spread changes, while a one-unit increase in lagged stock returns is associated with a 0.0037-unit decrease in next-day CDS spread changes [Table 4]. The economic significance and persistence of this trend is weak, though, as for both predictive variables, this relationship did not extend to the second day. Interestingly, we observe that when markets move into high EPU ranges, volatility is worse at predicting next-day spread differences, but stronger in the two-day lag.

Market uncertainty levels restructure the relationship and timing of the information channels between our three variables. We find that in the market cross-section, over the one-day lags, markets are absorbing shocks, whereas by the second day, the response reflects information reassessment [Table 3-4]. This is

reflected in the shift of significance from 1-day lags to 2-day lags from implied volatility to CDS spreads. The positive and significant predictive power of CDS spread on implied volatility indicates that a widening in default risk propagates into higher future volatility. Interestingly, the sign of this relationship, though still significant, flips under high policy uncertainty. The effect of CDS spread changes on our other variables seems inconclusive, as both the sign and direction of significance change when moving between lags and periods. The Wald tests for our market portfolio pass with strong significance in the directions of implied volatility to CDS spreads, Returns to CDS spreads and CDS to implied volatility across both regimes [Table 7]. The combination of joint significance across our structural model exemplifies that equity markets, options markets, and credit markets are jointly engaged in real-time price discovery.

7.2 Sector Portfolio Analysis

The second part of our empirical analysis refers to a deeper segmentation of the market into equal-weighted sector portfolios. In our results, we find that we can group our sectors empirically into three distinct categories [Table 9]. Firstly, sectors that become more tightly linked in volatility to CDS when moving into high EPU regimes: Consumer Goods, Consumer Services, Basic Materials, Utilities. In these sectors, we find that uncertainty shock events strengthen the information channel from implied volatility to CDS spreads. We hypothesize that under market stress, implied volatility is more precisely considered when pricing spreads, especially as increases in volatility imply higher default probabilities.

Secondly, sectors in which EPU regime changes do not largely affect the predictive relationships: Financials, Energy, and Industrials, though for the latter, CDS spreads gain predictive relevance for volatility under high EPU regimes. Thirdly, sectors in which CDS spreads lose autoregressive significance in the lag coefficients of the VAR when changing to high EPU Regimes: Healthcare, Financials, and Industrials [Table 10-12]. In these sectors, we find that high uncertainty periods reduce the persistence of CDS spreads, and previous day values have less of an effect on the future. To our knowledge, no previous literature has attempted to explain this phenomenon. Our results also reveal that across most sectors that have statistically significant autoregressive CDS spreads, the 1-day lag tends to have stronger economic significance than the 2-day, again implying weaker persistence. The substantial variety across our sectors signifies that although we can show systemic linkages between the variables across the market, at more refined levels, the level of white noise makes solid conclusions difficult.

Throughout our sectoral results, we can notice a common trend: Across both lags, CDS spreads produce more economically significant predictive effects on volatility and prices than in the reverse. This follows cross-sectional literature such as Galil (2014), which shows, although relevant, the low predictive power

of structural models overall in explaining CDS spread changes. Even with controls for additional firm and market characteristics, their models still only represent around 25% of the determinants of spread changes. Prices and implied volatility have more economically significant reactions to changes in CDS spreads, following intuitively as asset values react to an indicator of asset risk [Table 10-12].

Finally, we can interpret the endogenous thresholds allocated to each sector as they represent the point that minimizes the SSR and thus the level of policy uncertainty at which credit, options and stock price co-movement mechanisms change most sharply [Table 13]. The thresholds reveal substantial heterogeneity in the level of policy uncertainty required to trigger regime shifts across industries. Most sectors have thresholds in line with our market-wide threshold of 197.2. Though two substantial outliers emerge: Energy and Healthcare, whose regime thresholds are defined at 130.23 and 142.29, respectively. This suggests that even moderate increases in policy uncertainty are sufficient to trigger regime shifts in the behavior of their risk and valuation movements. We could make the argument that intuitively, Energy is strongly linked to geopolitical and economic shifts, while healthcare is highly exposed to regulatory or policy shifts and thus explain reactivity. However, because similar arguments could be made for other sectors, and there is a dearth of literature in this area, differences in industry exposure to EPU shocks remain a topic for future research.

8 Robustness Checks

Due to divergence in optimal lag length between AIC and Schwartz criterion tests, we also run Wald tests for market TVARs with a lag length of 10 days, holding trim constant at 0.15 as in our overall model, and find statistically significant relationships matching our 2-day lag results [Table 8]. However, for the direction of CDS spreads to volatility, we lose Granger causality, implying that this relationship may only have significant predictive power over shorter time horizons.

A trim of 0.15 was selected for our primary model based on Hansen’s (1996) recommendation of a trimming that follows [0.15, 0.85]. This trim size also allows us to keep at least 200 observations in the high EPU regime and presents a reasonable threshold value to divide the overall EPU trend [Figure 1]. As a robustness check on the size of our sample in each regime, we adjust the trims from 15% to separately allow for 10% and 20% of values in each regime [Table 7, Table 8]. We find mixed results in our robustness to threshold trimming. The second-lag predictive effect of implied volatility on CDS spread changes under high EPU is positive, economically large, and statistically significant across all trims. Likewise, widening CDS spreads consistently predict subsequent increases in implied volatility and stock price adjustments under high EPU. However, several relationships, especially in low uncertainty periods, seem to me more heavily dependent on TVAR trim size. Particularly, a trim of 0.1, allowing for fewer firms in the high uncertainty regime and

thus analysis at a higher threshold shows that 1-day lag CDS spreads have a significant effect on implied volatility, not present amongst the other trims. This sensitivity reflects the trade-off in threshold estimation between regime purity and sample size: smaller trims sharpen the definition of extreme uncertainty states at the cost of precision of estimation.

9 Conclusion

Through a time-series analysis of equal weighted market and sector portfolios we find strong evidence of omnidirectional Granger causality between our 3 primary variables: CDS spreads, implied volatility and stock price returns. By incorporating the EPU index as a proxy for overall market sentiment and macroeconomic shocks, we find that, at the market cross-section, high EPU regimes lead to shifts in the persistence of the autoregressive models and a reversal of sign in the CDS spread predictive power on implied volatility. When we decompose the market into sector portfolios, we continue to find frequent rejections of the null of no Granger causality (particularly from returns and implied volatility to CDS, and from CDS to volatility), but the associated VAR coefficients are unstable in sign, size and significance. Moreover, they are noisy across sectors, regimes and not-consistent with any financial economics models. Taken together, the results suggest that the market-wide linkage between credit risk, volatility, and stock returns is relatively robust, whereas sector-level dynamics are substantially more heterogeneous and dominated by idiosyncratic noise. Our findings are in line with prior evidence on the lack of stable determinants of CDS spreads across subsamples.

Another interpretation of our results speaks to the limitations of our methodology. Particularly, the error term in the VAR for sector level data could have a non-zero mean. The overall market and individual stock movements are largely noise, however sectors demonstrate specific behaviors over short to medium intervals of time. The results from the Portmanteau test also indicate that certain dynamics have been missed out while analysing sector level data for technology, industrials and financials. The thresholds for the Energy and Healthcare portfolio being significantly lower than the others also indicates that there are dynamics in the sector portfolios that our model and variables are not able to capture.

10 Appendix

Table 2: Market – Full Sample VAR (All Periods)

Equation	ΔCDS_{-1}	$\Delta\text{VOL}_{\text{imp},-1}$	ΔRET_{-1}	ΔCDS_{-2}	$\Delta\text{VOL}_{\text{imp},-2}$	ΔRET_{-2}
ΔCDS	0.1571***	0.0016**	-0.0009	0.0463	0.0049***	0.0013**
$\Delta\text{VOL}_{\text{imp}}$	1.9403	-0.0041	0.1556***	-0.0060	0.0446	0.0341
RET	-4.6875**	-0.0683*	-0.1592***	6.7659***	-0.1710***	-0.0742**

Table 3: Full-Sample Market Granger Causality Tests

Direction	F test	df1	df2	P value
Δ Volatility $\rightarrow$ Δ CDS	15.843***	4	4,107	7.11e-13
Δ Volatility $\rightarrow$ Returns	15.843***	4	4,107	7.11e-13
Returns $\rightarrow$ Δ CDS	12.637***	4	4,107	3.20e-10
Returns $\rightarrow$ Δ Volatility	12.637***	4	4,107	3.20e-10
Δ CDS $\rightarrow$ Returns	6.148***	4	4,107	6.27e-05
Δ CDS $\rightarrow$ Δ Volatility	6.148***	4	4,107	6.27e-05

Table 4: Threshold Vector Autoregressive Coefficients for Market Equal-Weighted Portfolio

Market – Low EPU (N = 1166)						
Equation	ΔCDS_{-1}	$\Delta\text{VOL}_{\text{imp},-1}$	RET_{-1}	ΔCDS_{-2}	$\Delta\text{VOL}_{\text{imp},-2}$	RET_{-2}
ΔCDS	-0.0748**	0.0018**	-0.0037***	0.1103***	0.0003	-0.0003
$\Delta\text{VOL}_{\text{imp}}$	0.1551	-0.2316***	-0.1044***	5.1802***	-0.1760***	-0.0097
RET	6.7214***	-0.0227	0.0253	-4.4772**	0.1275***	-0.0171
Market – High EPU (N = 210)						
Equation	ΔCDS_{-1}	$\Delta\text{VOL}_{\text{imp},-1}$	ΔRET_{-1}	ΔCDS_{-2}	$\Delta\text{VOL}_{\text{imp},-2}$	ΔRET_{-2}
ΔCDS	0.2214**	0.0003	-0.0045***	-0.1485*	0.0074***	-0.0019
$\Delta\text{VOL}_{\text{imp}}$	-6.4137	0.2570***	0.1642	-9.4312**	0.0779	-0.3677***
RET	-3.2937	-0.1058	-0.1885**	21.5209***	-0.3664***	0.1620*

Table 5: TVAR Market Coefficients (trim = 0.10, $\tau = 234.73$)

Market – Low EPU (N = 1225)						
Equation	ΔCDS_{-1}	$\Delta\text{VOL}_{\text{imp},-1}$	ΔRET_{-1}	ΔCDS_{-2}	$\Delta\text{VOL}_{\text{imp},-2}$	ΔRET_{-2}
ΔCDS	0.1517***	0.0011	-0.0006	-0.0137	-0.0004	-0.0013*
$\Delta\text{VOL}_{\text{imp}}$	7.5989***	-0.1858***	0.0551*	6.0287***	-0.1598***	-0.0066
RET	-6.8653***	-0.0128	-0.1767***	1.3124	0.1498***	0.0386
Market – High EPU (N = 151)						
Equation	ΔCDS_{-1}	$\Delta\text{VOL}_{\text{imp},-1}$	ΔRET_{-1}	ΔCDS_{-2}	$\Delta\text{VOL}_{\text{imp},-2}$	ΔRET_{-2}
ΔCDS	0.3192***	-0.0004	-0.0029	-0.2162**	0.0086***	0.0001
$\Delta\text{VOL}_{\text{imp}}$	-4.1260	0.2252**	0.2857**	-12.6098**	0.0453	-0.3236***
RET	-2.8077	-0.1471	-0.2931***	23.2987***	-0.4250***	0.1161

Table 6: TVAR Market Coefficients (trim = 0.20, $\tau = 172.45$)

Market – Low EPU (N = 1100)						
Equation	ΔCDS_{-1}	$\Delta\text{VOL}_{\text{imp},-1}$	ΔRET_{-1}	ΔCDS_{-2}	$\Delta\text{VOL}_{\text{imp},-2}$	ΔRET_{-2}
ΔCDS	0.0673*	0.0005	-0.0031***	0.1571***	-0.0024***	0.0003
$\Delta\text{VOL}_{\text{imp}}$	2.4625	-0.2278***	-0.0832***	6.3270***	-0.1561***	0.0209
RET	2.7792	-0.0761*	-0.0349	-6.3438***	0.0323	-0.0243

Market – High EPU (N = 276)						
Equation	ΔCDS_{-1}	$\Delta\text{VOL}_{\text{imp},-1}$	ΔRET_{-1}	ΔCDS_{-2}	$\Delta\text{VOL}_{\text{imp},-2}$	ΔRET_{-2}
ΔCDS	0.2036***	0.0006	-0.0006	-0.1224*	0.0087***	0.0009
$\Delta\text{VOL}_{\text{imp}}$	1.1621	0.1967**	0.3248***	-12.4417***	0.1424*	-0.1609*
RET	-6.6221*	-0.0499	-0.1372*	23.0177***	-0.3622***	0.0572

Table 7: Wald Test on Market Portfolio (Lag = 2)

Direction	P value (Low)	P value (High)
$\Delta\text{VOL}_{\text{imp}} \rightarrow \Delta\text{CDS}$	8.395e-04	2.335e-06
$\text{RET} \rightarrow \Delta\text{CDS}$	7.114e-07	1.754e-08
$\Delta\text{CDS} \rightarrow \Delta\text{VOL}_{\text{imp}}$	5.744e-05	4.721e-06

Table 8: Wald Test on Market Portfolio (Lag = 10)

Direction	P value (Low)	P value (High)
$\Delta\text{VOL}_{\text{imp}} \rightarrow \Delta\text{CDS}$	1.923e-11	8.454e-05
$\text{RET} \rightarrow \Delta\text{CDS}$	2.060e-09	2.537e-05
$\Delta\text{CDS} \rightarrow \Delta\text{VOL}_{\text{imp}}$	1.101e-02	3.265e-01

Figure 1: Economic Policy Uncertainty with Market TVAR Threshold (2017–2023)

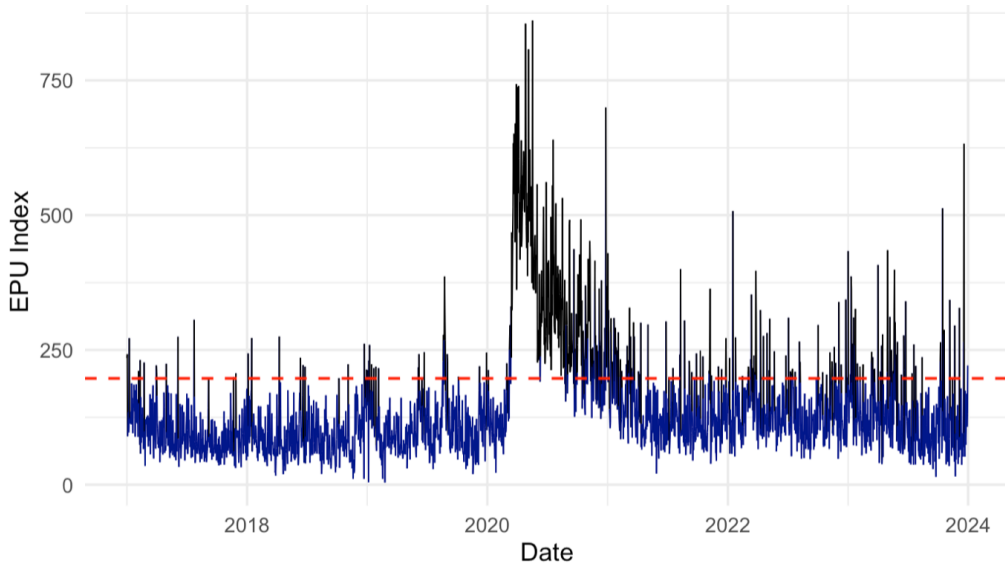


Table 9: Wald Tests on Sectoral Portfolios

Basic Materials		
Null Hypothesis	Low EPU	High EPU
CDS spreads do not Granger-cause implied volatility	0.000***	0.000***
Stock returns do not Granger-cause CDS spreads	0.000***	0.124
Implied volatility does not Granger-cause CDS spreads	0.108	0.097*
Consumer Goods		
CDS spreads do not Granger-cause implied volatility	0.000***	0.000***
Stock returns do not Granger-cause CDS spreads	0.000***	0.000***
Implied volatility does not Granger-cause CDS spreads	0.081*	0.000***
Consumer Services		
CDS spreads do not Granger-cause implied volatility	0.913	0.001***
Stock returns do not Granger-cause CDS spreads	0.000***	0.000***
Implied volatility does not Granger-cause CDS spreads	0.062*	0.000***
Energy		
CDS spreads do not Granger-cause implied volatility	0.000***	0.000***
Stock returns do not Granger-cause CDS spreads	0.000***	0.006***
Implied volatility does not Granger-cause CDS spreads	0.039**	0.288
Financials		
CDS spreads do not Granger-cause implied volatility	0.001***	0.037**
Stock returns do not Granger-cause CDS spreads	0.000***	0.003***
Implied volatility does not Granger-cause CDS spreads	0.010***	0.000***
Healthcare		
CDS spreads do not Granger-cause implied volatility	0.945	0.001***
Stock returns do not Granger-cause CDS spreads	0.000***	0.000***
Implied volatility does not Granger-cause CDS spreads	0.264	0.164
Industrials		
CDS spreads do not Granger-cause implied volatility	0.314	0.007***
Stock returns do not Granger-cause CDS spreads	0.000***	0.000***
Implied volatility does not Granger-cause CDS spreads	0.000***	0.001***
Technology		
CDS spreads do not Granger-cause implied volatility	0.421	0.002***
Stock returns do not Granger-cause CDS spreads	0.001***	0.000***
Implied volatility does not Granger-cause CDS spreads	0.012**	0.172
Utilities		
CDS spreads do not Granger-cause implied volatility	0.147	0.071*
Stock returns do not Granger-cause CDS spreads	0.005***	0.015**
Implied volatility does not Granger-cause CDS spreads	0.230	0.004***

Table 10: Vector Autoregressive Coefficients for Sectoral Equal-Weighted Portfolios (Part 1 of 3)

Basic Materials – Low EPU (N = 1166)						
Equation	ΔCDS_{-1}	ΔVOL_{-1}	ΔRET_{-1}	ΔCDS_{-2}	ΔVOL_{-2}	ΔRET_{-2}
ΔCDS	-0.0008	0.0004	-0.0034***	0.0523	0.0004	-0.0005
ΔVOL_{imp}	6.6896***	-0.4746***	-0.1204***	-0.0310	-0.2601***	-0.0545
RET	5.8764***	-0.0020	0.0534	-0.0791	0.0469*	0.0112
Basic Materials – High EPU (N = 210)						
ΔCDS	0.1213	0.0007	-0.0040**	-0.2632***	0.0025*	-0.0035*
ΔVOL_{imp}	-3.6421	-0.0589	0.0032	-9.5541**	0.0207	-0.1047
RET	-6.7752*	0.0083	-0.1602*	14.7555***	-0.1625***	0.0881
Consumer Goods – Low EPU (N = 1169)						
ΔCDS	-0.0289	-0.0008	-0.0050***	0.1296***	-0.0013*	0.0000
ΔVOL_{imp}	4.3125***	-0.4117***	-0.0744**	6.8578***	-0.1539***	0.0870***
RET	-0.4466	0.0595*	-0.0085	-2.9391**	0.0928***	-0.0818**
Consumer Goods – High EPU (N = 207)						
ΔCDS	0.2284***	0.0022	-0.0079***	-0.0136	0.0086***	-0.0021
ΔVOL_{imp}	-3.1131	-0.0658	0.0367	-11.5307***	0.2301***	-0.3038***
RET	-6.1298*	-0.1241**	-0.1469*	14.7648***	-0.3077***	0.0640
Consumer Services – Low EPU (N = 1167)						
ΔCDS	-0.2509***	0.0012**	-0.0051***	-0.0539*	0.0001	-0.0010
ΔVOL_{imp}	0.6301	-0.4824***	-0.2201***	1.0587	-0.2239***	-0.0531
RET	0.3184	0.0156	-0.0235	-0.8786	0.0397	-0.0377
Consumer Services – High EPU (N = 209)						
ΔCDS	-0.2721***	0.0040***	-0.0100***	-0.1793**	0.0059***	-0.0069***
ΔVOL_{imp}	-10.1944**	-0.0939	-0.3537***	-3.7543	0.1000	-0.3227***
RET	7.9750***	-0.1493***	0.0085	8.8721***	-0.0749	0.3374***

Table 11: Vector Autoregressive Coefficients for Sectoral Equal-Weighted Portfolios (Part 2 of 3)

Energy – Low EPU (N = 967)						
Equation	ΔCDS_{-1}	ΔVOL_{-1}	ΔRET_{-1}	ΔCDS_{-2}	ΔVOL_{-2}	ΔRET_{-2}
ΔCDS	0.2808***	-0.0015*	-0.0061***	-0.0035	-0.0015*	0.0012
ΔVOL_{imp}	11.0739***	-0.2840***	-0.1599***	3.2177*	-0.1440***	0.0453
RET	-0.6497	-0.0417	-0.0786**	-0.7552	-0.0557*	-0.0501
Energy – High EPU (N = 409)						
ΔCDS	0.2223***	-0.0001	0.0009	0.0037	0.0024*	-0.0061***
ΔVOL_{imp}	0.3546	-0.1530***	-0.0161	-2.6512	-0.0582	-0.1166
RET	5.0847**	-0.0171	-0.0573	5.5795***	-0.0795*	0.0525
Financials – Low EPU (N = 1166)						
ΔCDS	-0.1482***	0.0006*	-0.0030***	0.0213	-0.0002	-0.0008*
ΔVOL_{imp}	6.6951**	-0.3892***	-0.1536***	8.5485***	-0.2023***	-0.0183
RET	3.9163	-0.0171	-0.0222	0.2565	0.0690***	-0.0097
Financials – High EPU (N = 210)						
ΔCDS	0.0367	0.0018**	-0.0015	-0.0040	0.0024***	-0.0032***
ΔVOL_{imp}	-16.4120	0.0244	0.0837	1.9750	0.0561	-0.3132**
RET	14.7750*	-0.0815	-0.0057	11.2221	-0.2933***	0.0533
Healthcare – Low EPU (N = 881)						
ΔCDS	-0.3334***	0.0005	-0.0119***	-0.0663*	0.0004	-0.0053***
ΔVOL_{imp}	0.8092	-0.4467***	-0.1775***	0.2234	-0.1183***	0.0346
RET	0.0281	-0.0197	0.0456	0.2377	-0.0433**	-0.0256
Healthcare – High EPU (N = 495)						
ΔCDS	0.0666	0.0011	-0.0060***	0.0027	0.0018	-0.0020
ΔVOL_{imp}	3.4150**	-0.1528***	0.2240***	-1.2399	-0.1643***	-0.0027
RET	-2.1149**	-0.0357	-0.1515***	3.6726***	-0.0688**	0.0146

Table 12: Vector Autoregressive Coefficients for Sectoral Equal-Weighted Portfolios (Part 3 of 3)

Industrials – Low EPU (N = 1166)						
Equation	ΔCDS_{-1}	ΔVOL_{-1}	ΔRET_{-1}	ΔCDS_{-2}	ΔVOL_{-2}	ΔRET_{-2}
ΔCDS	-0.0423	0.0006	-0.0020***	-0.0640**	0.0015***	0.0001
$\Delta\text{VOL}_{\text{imp}}$	1.4297	-0.3865***	-0.1678***	0.7929	-0.2155***	-0.0719*
RET	2.4562	0.0591**	0.0379	-2.4679	0.0270	-0.0349
Industrials – High EPU (N = 210)						
ΔCDS	-0.0400	0.0015	-0.0037***	-0.1117	0.0026**	-0.0032**
$\Delta\text{VOL}_{\text{imp}}$	-0.8062	0.2266***	0.2159**	-4.8293	-0.0280	-0.3301***
RET	0.6867	-0.1143	-0.1086	15.5809***	-0.2538***	0.1152
Technology – Low EPU (N = 1126)						
ΔCDS	0.1653***	-0.0006	-0.0026***	0.1004***	-0.0005	-0.0005
$\Delta\text{VOL}_{\text{imp}}$	1.9437	-0.4681***	-0.1095**	3.0125	-0.2235***	-0.0335
RET	0.5614	-0.0498**	-0.0456	-1.1870	0.0135	0.0111
Technology – High EPU (N = 250)						
ΔCDS	0.1404**	0.0014	-0.0008	0.0750	0.0016	-0.0013
$\Delta\text{VOL}_{\text{imp}}$	-0.6505	-0.0336	0.3687***	-7.8693**	-0.0110	-0.1048
RET	-5.0579*	-0.0562	-0.1755**	9.4507***	-0.1259**	-0.0246
Utilities – Low EPU (N = 1156)						
ΔCDS	-0.0231	0.0003*	-0.0006**	-0.0623**	0.0004**	-0.0002
$\Delta\text{VOL}_{\text{imp}}$	4.1243	-0.5092***	-0.1527***	8.8281*	-0.1874***	-0.0617
RET	3.5455	-0.0093	0.0424	-4.3026	0.0075	0.0444
Utilities – High EPU (N = 220)						
ΔCDS	0.1745**	0.0005**	-0.0004	-0.1069	0.0008***	-0.0005
$\Delta\text{VOL}_{\text{imp}}$	50.9281**	-0.2012***	0.2895***	1.8085	0.0366	-0.0686
RET	-31.5486**	-0.0296	-0.2060***	0.2999	-0.1410***	-0.1033

Table 13: Endogenous Thresholds for Each Sector Portfolio

Sector	Threshold
Basic Materials	197.20
Consumer Goods	199.30
Consumer Services	198.18
Energy	142.29
Financials	197.20
Healthcare	130.23
Industrials	197.20
Technology	185.12
Utilities	192.35

Table 14: Eigenvalue Stability Test for Each Sector

Sector	Regime	N	Max Modulus	Stable
Basic Materials	Low EPU	493	0.487	TRUE
Basic Materials	High EPU	92	0.584	TRUE
Consumer Goods	Low EPU	477	0.390	TRUE
Consumer Goods	High EPU	108	0.747	TRUE
Consumer Services	Low EPU	446	0.517	TRUE
Consumer Services	High EPU	139	0.532	TRUE
Energy	Low EPU	493	0.518	TRUE
Energy	High EPU	92	0.651	TRUE
Financials	Low EPU	497	0.417	TRUE
Financials	High EPU	88	0.655	TRUE
Healthcare	Low EPU	494	0.544	TRUE
Healthcare	High EPU	91	0.580	TRUE
Industrials	Low EPU	497	0.490	TRUE
Industrials	High EPU	88	0.598	TRUE
Technology	Low EPU	451	0.454	TRUE
Technology	High EPU	134	0.642	TRUE
Utilities	Low EPU	495	0.394	TRUE
Utilities	High EPU	90	0.455	TRUE

Table 15: Joint Portmanteau Test Results

Market Portfolio			
Regime		N	P value
Low EPU		1100	2.82e-06
High EPU		276	6.08e-10
Sectoral Portfolios			
Sector	Regime	N	P value
Basic Materials	Low EPU	1160	2.67e-02
Basic Materials	High EPU	208	2.68e-07
Consumer Goods	Low EPU	1148	8.02e-03
Consumer Goods	High EPU	220	4.30e-09
Consumer Services	Low EPU	1162	3.53e-03
Consumer Services	High EPU	206	1.16e-05
Energy	Low EPU	1148	7.73e-11
Energy	High EPU	220	1.17e-03
Financials	Low EPU	1162	1.64e-01
Financials	High EPU	206	3.35e-04
Healthcare	Low EPU	1156	5.52e-07
Healthcare	High EPU	212	1.07e-02
Industrials	Low EPU	1157	4.07e-02
Industrials	High EPU	211	5.62e-02
Technology	Low EPU	1118	3.27e-05
Technology	High EPU	250	5.16e-02
Utilities	Low EPU	1157	6.47e-03
Utilities	High EPU	211	3.42e-03

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